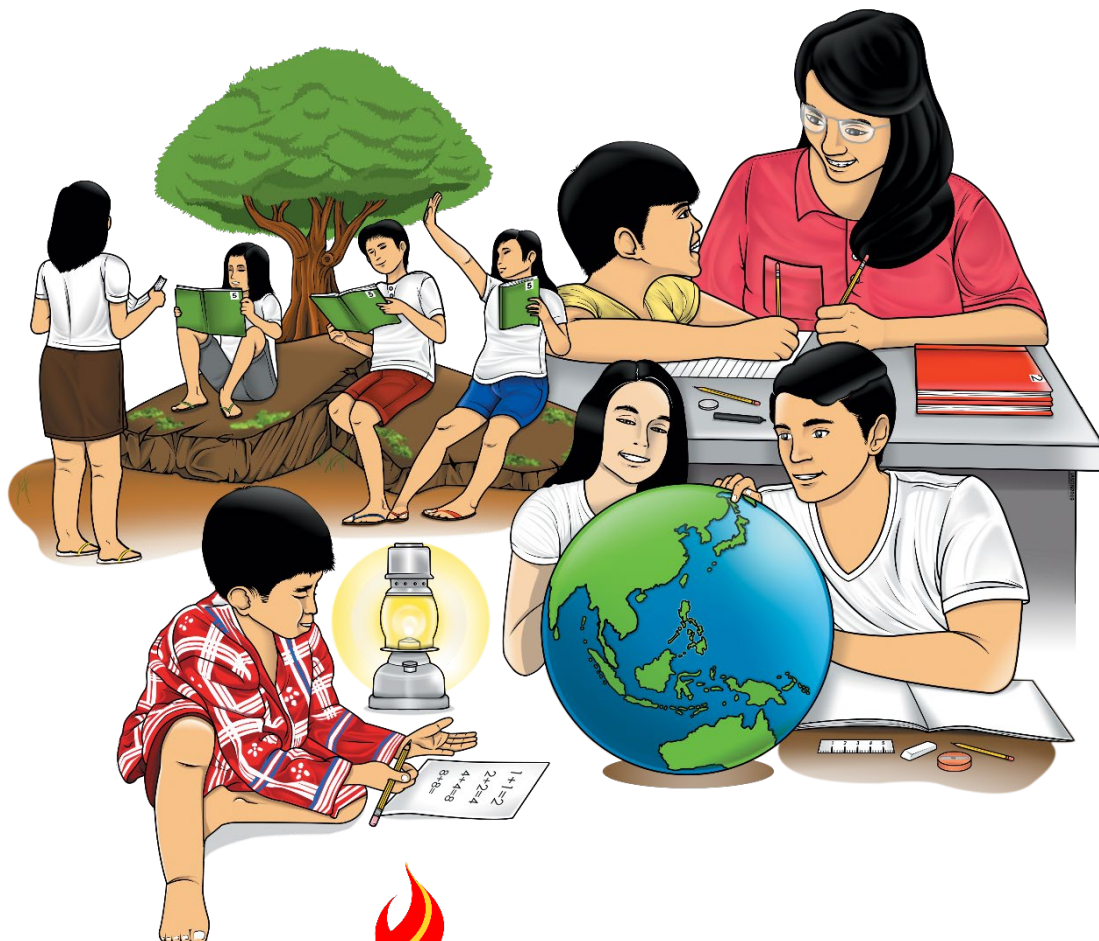


Science

Quarter 2 – Module 7

Protection and Conservation of Ecosystem



Science – Grade 6
Alternative Delivery Mode
Quarter 2 – Module 7: Protection and Conservation of Ecosystem
First Edition, 2020

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Science

Quarter 2 – Module 7: Protection and Conservation of Ecosystem

Introductory Message

This Self-Learning Module (SLM) is prepared so that you, our dear learners, can continue your studies and learn while at home. Activities, questions, directions, exercises, and discussions are carefully stated for you to understand each lesson.

Each SLM is composed of different parts. Each part shall guide you step-by-step as you discover and understand the lesson prepared for you.

Pre-tests are provided to measure your prior knowledge on lessons in each SLM. This will tell you if you need to proceed on completing this module or if you need to ask your facilitator or your teacher's assistance for better understanding of the lesson. At the end of each module, you need to answer the post-test to self-check your learning. Answer keys are provided for each activity and test. We trust that you will be honest in using these.

In addition to the material in the main text, Notes to the Teacher are also provided to our facilitators and parents for strategies and reminders on how they can best help you on your home-based learning.

Please use this module with care. Do not put unnecessary marks on any part of this SLM. Use a separate sheet of paper in answering the exercises and tests. And read the instructions carefully before performing each task.

If you have any questions in using this SLM or any difficulty in answering the tasks in this module, do not hesitate to consult your teacher or facilitator.

Thank you.



What I Need to Know

This module was designed and written with you in mind. It is here to help you master the ways on how to protect and conserve the tropical rainforests, coral reefs and mangrove swamps. The scope of this module is used in many different learning situations. The language used recognizes the diverse vocabulary level of students. The lessons are arranged to follow the standard sequence of the course. But the order in which you read them can be changed to correspond with the textbook you are now using.

The module is about:

- Identifying ways to protect and conserve the tropical rainforest, coral reefs and mangrove swamps
- Explaining the need to protect and conserve tropical rainforests, coral reefs and mangrove swamps.

After going through this module, you are expected to:

- Identify ways to protect and conserve the tropical rainforest, coral reefs and mangrove swamps
- Explain the need to protect and conserve tropical rainforests, coral reefs and mangrove swamps.



What I Know

Directions: Choose the letter of the best answer. Write your answer in your science journal/notebook.

1. Which statement best describes conservation?
 - A. use of natural resources
 - B. maintaining the life of ecosystem
 - C. wise use and preservation of resources
 - D. utilizing of earth's resources
2. Which human activity contributes to the destruction of forest?
 - A. reforestation
 - B. deforestation
 - C. protecting the forest against harmful activities
 - D. all of the above
3. The following are ways to conserve Earth's resources, which one is **NOT**?
 - A. proper waste disposal
 - B. reduce air and water pollution
 - C. buy products made from animals
 - D. help raise funds for environmental conservation
4. Which of the following destroys coral reefs?
 - A. building man-made fish sanctuaries
 - B. dynamite fishing
 - C. waste dumping
 - D. both B and C
5. Ecosystem is an environment where both living and non-living things exist and _____ with one another.
 - A. disagree
 - B. interact
 - C. compete
 - D. disconnected
6. Which is a cause of habitat destruction?
 - A. too many plants
 - B. thriving animals
 - C. great weather
 - D. pollution

7. Species that are in danger of going extinct are called?
- A. endangered
 - B. threatened
 - C. fossilized
 - D. adapted
8. Which situation does not maintain biodiversity?
- A. Growing plants on a farm
 - B. Protecting all the species in an ecosystem
 - C. Protecting the habitats in an ecosystem
 - D. Removing an important predator from the ecosystem
9. When we cut down trees, what do we do to the habitats of many living things?
- A. help
 - B. destroy
 - C. build
 - D. plant
10. What is/are the way/ways to restore denuded forest?
- A. reforestation
 - B. deforestation
 - C. *kaingin* system
 - D. dynamite fishing

Lesson

1

Protection and Conservation of Ecosystems

Forest, coral reefs, and mangrove swamp ecosystems need to be protected and conserved. We need to give importance to these ecosystems because they provide us with our needs.



What's In

Directions: Write **PC** if the following statement shows protecting and conserving tropical rainforest, coral reefs and mangrove swamps and write **N** if not. Write your answer in your Science Journal.

- _____ 1. Replanting trees in the forest.
- _____ 2. Striking or touching a coral reef.
- _____ 3. Helping organizations dedicated in protecting and conserving natural resources.
- _____ 4. Saying no to deforestation.
- _____ 5. Establishing park for recreation.
- _____ 6. Maintaining sustainable limits in logging.
- _____ 7. Participating in local coastal cleanups.
- _____ 8. Recycling and disposing trash properly.
- _____ 9. Clearing mangroves for property development.
- _____ 10. Restoring mangrove forest.



What's New

The Ecosystem is an environment where both living and non-living things exist and interact with one another. Identify what type of ecosystem is shown in each picture.

1.



2.



3.





What is It

The ecosystem is an environment where both living and non-living things exist and interact with one another. Examples of ecosystem are the tropical rainforests, coral reefs, and mangrove swamps.

A tropical rainforest is simply an area where tall, mostly evergreen trees grow and with high amount of rainfall.

Coral reefs serve as habitat for many animals. They are breeding ground of marine life. They are composed of non-living components such as water and sand and living components such as fish, crustaceans, mollusks, cnidarians, sponges and echinoderms.

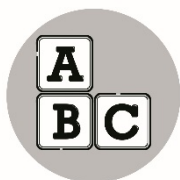
Mangrove swamp ecosystem is composed mostly of mangrove plants and animals like crustaceans and migratory birds. It is an important system that allows for the breeding of fishes and survival of other marine animals. It is also a part of the coastal and marine ecosystems.

Today, tropical rain forests, coral reefs and mangrove swamps are disappearing from the face of the globe. Despite growing international concern, these are continuing to be destroyed. These destructions have a global impact through species extinction, the loss of important ecosystem services and resources. However, these destructions can be slowed, stopped and in some cases even reversed.

We need to protect and conserve the Earth's resources. We need to give high value to our tropical rain forests, coral reefs and mangrove swamps. Here are some ways to protect and conserve these natural resources.

1. Habitat restoration
2. Reforestation or replanting a forest
3. Mangrove rehabilitation.
4. Wildlife conservation which includes protecting a sanctuary where animals live.
5. Supporting projects of the government and non-government organization that protect tropical rainforest, coral reefs and mangrove swamps.
6. Joining organizations that conduct educational campaigns and give information on how to protect and conserve our ecosystem.

Being responsible to the development and restoration of Earth's resources like the tropical rainforests, mangrove swamps and coral reefs is a key for the long-term benefits to mankind.



What's More

Directions: Read and understand the following statements. Write your answer in your science journal.

1. Give at least 2 ways on how to protect and conserve living organisms found in the tropical rainforest, coral reefs and mangrove swamps.

2. Explain why we need to protect and conserve the following: tropical rainforest, coral reefs and mangrove swamps of the Philippines.

tropical rainforest:

coral reefs:

mangrove swamps:



What I Have Learned

Directions: Complete the statement below. Write your answer in your science journal.

I have learned that...

Tropical rainforests, coral reefs and mangrove swamps are some examples of _____. Hence, they should be protected and conserved. There are some ways to protect and conserve our ecosystem like habitat restoration, _____, _____, _____ and wildlife conservation.



What I Can Do

Directions: Read and write your ideas on the following. Write your answer in your Science Journal.

1. Provide at least 3 major threats to tropical forest?

2. Mangrove forest had been classified by many governments and industries alike as wastelands, or useless swamps. Do you agree with that? Why?

3. The coral reefs around the world face threats from climate change, overfishing and pollution. As a grade VI pupil, how can you help protect or lessen the threats for the coral reefs?

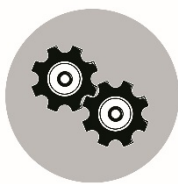


Assessment

Directions: Choose the letter of the best answer. Use a separate sheet for your answer.

1. Which of the following destroys coral reefs?
 - A. mangrove planting
 - B. building man-made fish sanctuaries
 - C. waste dumping
 - D. coastal clean up
2. Which of the following strategies show how to conserve the balance of life in the ecosystem?
 - A. massive cutting of trees
 - B. conserving the needs of living things
 - C. destroying wildlife ecosystem
 - D. dumping garbage anywhere

3. Which of the following shows wildlife conservation?
 - A. excessive hunting of wild animals
 - B. protecting the forests against harmful activities
 - C. practicing *kaingin* system in the forest
 - D. muro-ami fishing
4. The following shows mangrove rehabilitation, EXCEPT:
 - A. Planting mangrove seedlings in the mangrove swamps.
 - B. Supporting projects for mangrove rehabilitation.
 - C. Making the mangrove swamps as dumping site.
 - D. Joining the campaign against mangrove destruction.
5. The _____ is one of the ways to restore denuded forest.
 - A. reforestation
 - B. deforestation
 - C. *kaingin* system
 - D. dynamite fishing
6. Why is dynamite fishing being prohibited?
 - A. The use of explosives contributes food to marine animals.
 - B. The use of explosives benefits plants and animals.
 - C. The use of explosives improves marine sanctuary.
 - D. The use of explosives destroys the marine life.
7. Massive _____ will destruct natural habitat of wildlife.
 - A. reforestation
 - B. reclamation
 - C. deforestation
 - D. conservation
8. Why should mangroves be conserved and protected?
 - A. The mangrove area makes coastal erosion worse.
 - B. The mangroves provide shelter for fishes and crustaceans.
 - C. The mangroves serve as dumping site.
 - D. The mangroves serve as area for fishing.
9. Here are some ways of protecting and conserving wildlife, which one is not?
 - A. Buy products made from animal parts.
 - B. Protect a fish sanctuary.
 - C. Support programs for wildlife conservation.
 - D. Help raise funds for environment conservation.
10. What areas or places where fishing and hunting are prohibited?
 - A. rivers
 - B. swamps
 - C. coastal
 - D. sanctuaries



Additional Activities

Directions: Make a poster showing how to protect and conserve the tropical rainforests, coral reefs and mangrove swamps.

RUBRICS FOR POSTER MAKING

Category	10 pts.	8 pts.	6 pts.	4 pts.
Coverage of the topic	The poster captures the important information about the topic	The poster captures the important information about the topic but need more information to fully understand it.	The poster needs more information to further understand it.	The poster has nothing to do with the topic
Use of Graphics	All graphics are related to the topic	All graphics are related to the topic, but some are vague	1 graphics is not related to the topic	2 or more graphics are nor related to the topic
Organization	All information are organized well	1 information is not organized well	2 information are not organized well	3 or more information are not organized well
Layout & Design	All information on the poster can be clearly identified	Most of the information on the poster can be clearly identified	Some of the information on the poster can be clearly identified	Most of the information on the poster are not easy to understand
Mechanics	No spelling or punctuation error	Almost no spelling or punctuation error	A few spelling and punctuation error	Many spelling and punctuation error



Answer Key

Assessment 1. C 2. B 3. B 4. C 5. A 6. D 7. C 8. B 9. A 10. D Additional Activities Posters may vary What I Know 1. C 2. B 3. C 4. D 5. B 6. D 7. A 8. D 9. B 10. A	What I Have Learned ecosystem reforestation, mangrove rehabilitation, supporting projects of the government and non-government organization, joining organization that conduct educational campaign on how to protect and conserve tropical rainforest, mangrove swamp and coral reefs What Can I Do 1. Human threats are the major threats in tropical rainforest. 2. No. Mangrove swamps are important because they protect the shorelines from damaging storms, waves and floods. 3. By not throwing garbage that can pollute the water and can harm the reef. (Answers may vary)	What's In 1. PC 2. N 3. PC 4. PC 5. N 6. PC 7. PC 8. PC 9. N 10. PC What's New 1. coral reef 2. tropical rainforest 3. mangrove swamp What's More Answers may vary
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